



QKD Iterative Information Reconciliation Based on LDPC Codes

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Abstract

In this paper, we discuss the problem of QKD iterative information reconciliation based on LDPC codes. The aim is to correct high error rates in longer distances or in the condition of noisier environments. We use the methods of data reordering, data grouping, bit filling and error groups iterative reconciliation to improve the reconciliation capability and enhance reconciliation data security, and it can also reduce the impact of burst errors and retain more useful raw data. The simulation experiments have shown that the error-correcting ability of the iterative information reconciliation scheme proposed in this paper is beyond the results of traditional one-way scheme.

Keywords Quantum key distribution · Information reconciliation · Low-density parity-check codes

1 Introduction

Quantum key distribution (QKD) allows two distant parties to share secret keys safely. Traditionally, Alice sends qubits message, Bob receives the qubits, and Eve is the malevolent eavesdropper. QKD overcomes the flaws of conventional cryptography. It is unconditional security, which is guaranteed by the fundamental laws of quantum physics rather than assumptions of the eavesdropper's computational power. The laws of quantum physics include the Heisenberg's uncertainty principle [1] and quantum no-cloning theorem [2].

According to the QKD protocol, such as BB84, after Bob measures these quantum states that Alice sent over the quantum channel, then Alice and Bob exchange information over the classical channel to achieve their sifted key to work with. But the key generated by

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this protocol may contain some errors due to the imperfections of devices and quantum channel or any the attack of the eavesdroppers on the quantum channel, and noise in the quantum channel also introduces unavoidable errors in Bob's measures [3]. According to the free space experiment in [4], from a total quantum bit error rate (QBER) of 9.9%, the Vienna group attributes 1.5% to dark counts, 3.5% to source imperfections, and the rest to polarisation misalignments. To overcome this, one can correct errors by using information reconciliation methods. Information reconciliation is one of the methods to find and remove the erroneous bits and make them identical between Alice's and Bob's sifted keys. This method is based on error-correcting codes and done through classical open channels. The final secret key rate and maximal transmission distance of QKD systems are both affected by the ability of information reconciliation [5, 6].

An optimal information reconciliation method should have three characteristics: minimizing the information leakage to eavesdropper to ensure its safety, trying to reduce the bit error rate and improve its usability and trying to retain more useful raw data to ensure its reliability. Information reconciliation for QKD is a problem already studied in many literatures. The most widely used methods are the binary algorithm, the Cascade algorithm, the Winnow algorithm, and the low-density parity-check (LDPC) codes based method [7]. The binary algorithm is a reconciliation protocol that detects and corrects discrepancies by binary search and parity check [8, 9]. The advantages of this algorithm lie in its high efficiency of processing and low computational complexity, but each round can only correct one error bit and cannot correct even errors. So the algorithm requires multiple rounds of information interaction. The Cascade algorithm [10] solves the problem that even errors cannot be detected by the binary algorithm. Its efficiency is not optimal [11] because of its highly interactive. Some modified versions have been proposed in literatures, which try to improve it either by optimizing the parameters in the algorithm or by modifying the protocol itself [12–14]. The aim is to improve the reconciliation efficiency, or reduce the needed number of communication rounds. The Winnow algorithm [14] is based on Hamming codes and is a modified versions of Cascade, which can solve the problems of bounding information leakage and the extremely high communication complexity [8]. After exchanging the parities of every block, Alice and Bob exchange the syndrome of a Hamming code for correcting single error in each block with a parity mismatch [12]. The Winnow introduces additional errors to multiple errors block because the Hamming algorithm can only reveal one single error in each block [7]. The LDPC codes based method has the ability to correct the same range of errors as Cascade and has the advantage to improve the safety of the used protocol due to less information leakage [7], it can also reduce the communication delays compared to the Cascade algorithm [15]. Its channel coding and decoding are based on syndrome S , and it is a one-way protocol [8], the uncorrected blocks will be discarded.

Now QKD can be successfully used over longer distances or in noisier environments, QBER increases with the increasing of transmission distance [16], which indicates that the information reconciliation method must be more efficient in these situations. In this paper we describe an iterative information reconciliation protocol based on LDPC codes to adapt to high error rates for QKD system over longer distances or in the condition of noisier environments. Our method helps to further ensure the security of data transmission by data reordering and bit filling, and helps to further improve the error correction capability for high QBER over long distance, and retains more useful raw data by error groups iterative reconciliation. Furthermore the data reordering can reduce the impact of burst errors. The proposed method consists of six procedures: data reordering, data grouping, bit filling,

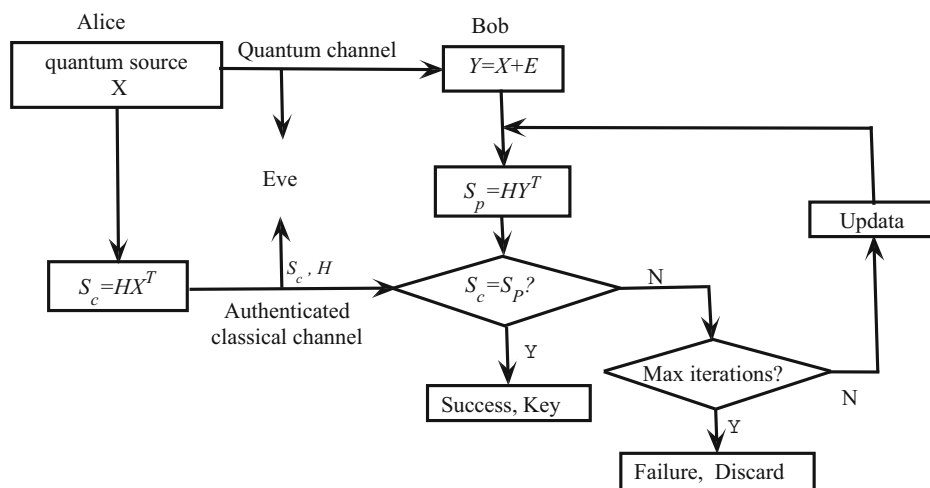


Fig. 1 The system model of one-way information reconciliation scheme based on LDPC codes

generating side information and sending by Alice over the classical authenticated channel, joint decode and error groups iterative reconciliation. The rest of this paper is organized as follows: the basic knowledge of information reconciliation for QKD and the principle of LDPC codes are introduced in Section 2. Our proposed iterative information reconciliation method based on LDPC codes is described in Section 3. In Section 4 we show some simulation results of our iterative information reconciliation scheme. Section 5 gives a conclusion.

2 QKD Information Reconciliation Based on LDPC Codes

2.1 Information Reconciliation Model

The QKD includes discrete-variable (DV)QKD [17–19] and continuous-variable (CV) QKD [20, 21]. DVQKD is based on discrete variables, such as the polarization of a single photon, both sender and receiver are binary data, linear block code algorithm can be used as reconciliation algorithm directly. In CVQKD, the information is encoded on the continuous variables, such as the position or momentum quadratures of coherent states. The proposed information reconciliation method is based on DVQKD. While for the CVQKD, quantizing and binary encoding of continuous variables are required [22]. Information reconciliation is to ensure that two distance parties who have two correlated sequences X and Y to share an identical sifted key with each other through error correction coding. As stated early in many literatures information reconciliation can be viewed as a special case of error correction with side information at the decoder studied by Slepian and Wolf [23]. Let $X = (x_1 x_2 \cdots x_n)$ and $Y = (y_1 y_2 \cdots y_n)$ are the initial key sequences, $x_i, y_i \in \{0, 1\}$ are the values of X and Y respectively. The system model of information reconciliation is shown in Fig. 1.

First Alice and Bob obtain the initial key sequences X and Y respectively after quantum transmission and channel parameter estimation, where we assume X and Y are both binary information sequences, sequence X is the standard key, and the error pattern between sequence Y and sequence X is E , that is, $Y = X + E$. The encoder uses X to generate a

compressed data S of some fixed length. Data S are transmitted via authenticated classical channel, then at receiver, using both S and Y , Bob jointly decodes in order to correct the errors in his data to match Alice's by using information reconciliation algorithm. Later, Alice and Bob confirm that the reconciliation is successful.

According to the direction of data transmission through classical open channels, information reconciliation algorithm can be divided into direct reconciliation and reverse reconciliation. In direct reconciliation the classical information flow has the same direction as the initial quantum information flow. That is Alice sends correction information and Bob corrects his key elements to have the same values as Alice. It will provide a useable secret key if $I_{AB} - I_{AE} > 0$ at the beginning, where I_{AB} is the mutual information between Alice and Bob, I_{AE} is the mutual information between Alice and the potential eavesdropper Eve [24]. In reverse reconciliation Bob sends correction information to Alice and Alice corrects her key elements to have the same values as Bob. This type of reconciliation will provide a useable key if $I_{AB} - I_{BE} > 0$, where I_{BE} is the mutual information between Bob and Eve.

2.2 LDPC Codes

LDPC codes are linear error-correcting codes with a sparse parity-check matrix. LDPC codes are essential to achieve performance close to the Shannon capacity limit. It can be completely determined by its parity-check matrix H . LDPC codes may also be represented by bipartite graphs that are directly related to their parity-check matrices H , one set of the nodes are check nodes representing the vertices for the parity-check equations; the other are variable nodes (or bit nodes) which represent the vertices for the codeword bits. Variable nodes and check nodes are connected through edges, the bipartite graph is used in the message-passing decoding algorithm. LDPC codes can be divided into regular and irregular LDPC codes. For regular LDPC codes, all degrees of the variable nodes are equal and also all degrees of the check nodes are equal. For irregular codes each variable node and each check node can have different degrees assigned, irregular LDPC codes can be characterized by the degree distribution polynomials $\lambda(x)$ and $\rho(x)$:

$$\lambda(x) := \sum_{i=2}^{d_v} \lambda_i x^{i-1} \quad (1)$$

$$\rho(x) := \sum_{i=2}^{d_c} \rho_i x^{i-1} \quad (2)$$

where λ_i is the fraction of edges in the bipartite graph connected to degree- i variable nodes, ρ_i is the fraction of edges connected to degree- i check nodes, and $\sum_i \lambda_i = \sum_i \rho_i = 1$. d_v and d_c denote the maximum degree for variable nodes and check nodes, respectively. The degree distribution pair can be used to predict a decoding threshold for the LDPC codes [25].

2.3 Encoding Scheme of LDPC for Information Reconciliation

For the giving binary sequence $X = (x_1 x_2 \cdots x_n)$ of length n , we form the syndrome bits S as an encoder output by multiplying the source bits X with the parity-check matrix H such that

$$S = \text{mod}\{H * X^T, 2\} \quad (3)$$

where $S = (s_1 s_2 \cdots s_{n-k})$, $s_j \in \{0, 1\}$ is the value of the j^{th} syndrome component. Equivalently, in the bipartite graph, this can be viewed as binary addition of all the variable node values that are connected to the same check node. H is a $(n - k) \times n$ LDPC parity-check matrix. Since H matrix is sparse, the encoding complexity of LDPC codes for traditional error correction applications is the square of the length of codeword.

The encoded version of X is the syndrome S which is the input to the authenticated classical channel. The source Y is available at the decoder as side information. Using a linear (n, k) binary block code, it is possible to have 2^{n-k} distinct syndromes, each indexing a set of 2^k binary words of length n [26]. This encoding results in mapping a sequence of n input symbols into $(n - k)$ syndrome symbols.

2.4 Sum-product Decoding for Information Reconciliation

With the side information Y and the receiving syndrome S , the goal of decoder is to reconstruct the best estimate $\hat{X}$ of the source n -sequence X . The basic LDPC decode algorithm includes bit-flipping algorithm and sum-product algorithm. The bit-flipping algorithm is based on the principal that a codeword bit involved in a large number of incorrect check equations is likely to be incorrect itself. So if the majority of the messages received by a bit node are different from its received value the bit node changes (flips) its current value.

Iterative decoding of binary LDPC codes using the sum-product algorithm (SPA) is a soft decision message-passing algorithm. The input bit probabilities are called a priori probabilities for the received bits, the bit probabilities returned by the decoder are called a posteriori probabilities. In the case of sum-product decoding these probabilities are expressed as log-likelihood ratios (LLR). The decoding steps of information reconciliation for QKD similar to the classical belief-propagation algorithm except the modification at the following step 2.

The LLR of a binary random variable x is defined as:

$$L(x) = \log_e \frac{p(x=0)}{p(x=1)} \quad (4)$$

where $p(x=1)$ denotes the probability that the random variable x takes the value 1, and

$$p(x=1) = \frac{\frac{p(x=1)}{p(x=0)}}{1 + \frac{p(x=0)}{p(x=1)}} = \frac{e^{-L(x)}}{1 + e^{-L(x)}} \quad (5)$$

For a binary variable x , it is easy to find $p(x=0)$ given $p(x=1)$, since $p(x=1) = 1 - p(x=0)$. The LLR-SPA is summarized as follows:

Step 1: Initialization: Set the log-likelihood ratios p to proper initial values based on the knowledge of the mean source correlation. i.e.,

$$M_{j,i} = r_i \quad (6)$$

where $M_{j,i}$ is the messages sent from the bit nodes to the check nodes, $i = 1, 2, \dots, n$, $j = 1, 2, \dots, n - k$, r_i is the log likelihood ratios for the priori message probabilities in BSC, and

$$r_i = \begin{cases} \log_e \frac{p}{1-p}, & \text{if } y_i = 1 \\ \log_e \frac{1-p}{p}, & \text{if } y_i = 0 \end{cases} \quad (7)$$

Step 2: Calculate $E_{j,i}$: the extrinsic message from check node j to bit node i , it is the LLR of the probability that bit i causes parity-check j to be satisfied.

$$E_{j,i} = \log_e \left(\frac{1 + (1 - 2s_j) \prod_{i' \in B_j, i' \neq i} \tanh(M_{j,i'}/2)}{1 - (1 - 2s_j) \prod_{i' \in B_j, i' \neq i} \tanh(M_{j,i'}/2)} \right) \quad (8)$$

where B_j is the set of bits in the j^{th} parity-check equation of the code. s_j is the j^{th} syndrome component sent from Alice. Equation (8) is the only modifications to the classical belief-propagation algorithm, where information is propagated between variable and check nodes that are connected by edges.

Step 3: Calculate the total LLR of the i^{th} bit by sum of these LLRS:

$$L_i = \sum_{j \in A_i} E_{j,i} + r_i, \quad i = 1, 2, \dots, n \quad (9)$$

where A_i is the parity-check equations which check on the i^{th} bit of the code,

Step 4: Hard decision on the received bits given by the sign of the LLRs.

$$z_i = \begin{cases} 1, & \text{if } L_i \leq 0 \\ 0, & \text{if } L_i > 0 \end{cases} \quad (10)$$

Step 5: Check if $Z = (z_1 z_2 \dots z_n)$ is a valid codeword,

$$S' = \text{mod}\{H * Z^T, 2\} \quad (11)$$

Then Bob compares the syndrome S' with the syndrome S sent from Alice. If $S' \neq S$, then there are errors in Y , and go back to step2, the belief propagation of LDPC decode is iteratively computed until $S' = S$, or the maximum allowed iteration number of decoding is reached. The result $S' = S$ means that the sifted key is identical in Alice's and Bob's after information reconciliation, otherwise there are still some erroneous bits between Alice's and Bob's sifted keys. These erroneous bits can no longer be corrected in one-way reconciliation protocols.

3 The Iterative Information Reconciliation Based on LDPC Codes

One problem in applying LDPC codes is building the parity-check matrix H , there a number of papers that have been published to discuss it [27, 28], such as the IEEE 802.11n Std [15], IEEE 802.16e Std [16] and ETSI DVB Std [3], these matrices have a systematic structure. In this paper, we use IEEE 802.11n standard to build the parity-check matrix H . According to the IEEE 802.11n standard, the code rates can be 1/2, 2/3, 3/4 or 5/6, the codeword length can be 648, 1296, or 1944 et al..

Here, a simulation is made to analysis the one-way information reconciliation capability of $H_{324 \times 648}$ and $H_{972 \times 1944}$ respectively, where the code rate is 1/2, the length of sifted key is 100000, and the maximum iterations number of the LDPC decoder is set to 100, the sum-product decoding algorithm is used to decode the LDPC codes at receiver. Figure 5 shows QBER before and after information reconciliation. Figure 6 shows the relationship between initial QBER and frame error rate (FER), FER is defined as the ratio of uncorrected number of groups after reconciliation to the total number of groups. We can see that the remaining QBER decreases to 0 after information reconciliation by using parity-check matrix $H_{972 \times 1944}$ when the initial QBER is below 5%, and the remaining QBER decreases

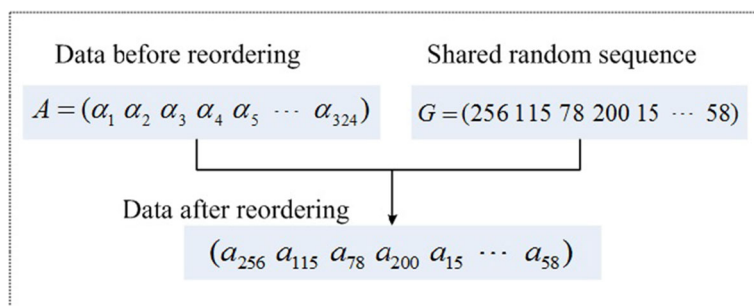


Fig. 2 Data reordering of sequence A according to G

to 0 after information reconciliation by using parity-check matrix $H_{324 \times 648}$ when the initial QBER is below 4%.

In QKD, the error rates over 11% are considered unacceptable for BB84, and the eavesdroppers are considered to exist and the communication will be discarded. If the QBER is less than 11%, Alice and Bob can form a shared key with minimal information known to an eavesdropper [29], by classical error correction and privacy amplification [30].

From Fig. 5, We can see if the QBER is above 5%, the error correction capability will be greatly reduced in one-way LDPC reconciliation scheme. So we proposed an iterative information reconciliation scheme in order to increase the error correction capability over longer distances or in noisier environment. The iterative information reconciliation scheme uses data reordering, data grouping, bit filling and information reconciliation again of the uncorrected groups to improve the reconciliation capability. The details are as follows.

Let us suppose that the binary sequence sent by Alice is denoted as $A = (a_1 a_2 \dots a_N)$, the binary sequence received by Bob is denoted as $B = (b_1 b_2 \dots b_N)$, where N is the length of the binary information sequence.

Data reordering: First Alice and Bob reorder their data according to a shared random sequence (either part of their initially shared sequence, or communicated on the classical channel) in order to make the errors uniformly random distributing to reduce the impact of burst errors. For example, assume the shared random sequence is $G = (g_1 g_2 \dots g_N)$, Alice and Bob reorder their binary sequences according to sequence G , then, Alice has a N -bit binary string $A = (\alpha_1 \alpha_2 \dots \alpha_N)$, Bob has $B = (\beta_1 \beta_2 \dots \beta_N)$. Here $G = (g_1 g_2 \dots g_N)$ is a vector containing N unique integers selected randomly from 1 to N . For example, $G = (g_1 g_2 \dots g_7)$ might be (4 6 1 7 2 5 3). The operation of A and G is to reorder the element position in A according to the correspondent element value in G . Figure 2 shows an example.

Data grouping: Data grouping is that both Alice and Bob divide their binary sequence to some equal length groups. The length of each group depends on the LDPC parity-check matrix H , since H is a $(n - k) \times n$ matrix, the length of groups should be n , but in order to increase the error correction capability of the LDPC codes, the length of each group is less than n . Then after data grouping, the following groups are obtained: $(\alpha_1 \alpha_2 \dots \alpha_\mu)$, $(\alpha_{\mu+1} \alpha_{\mu+2} \dots \alpha_{2\mu})$, $\dots$, $(\alpha_{(k-1)\mu+1} \alpha_{(k-1)\mu+2} \dots \alpha_{k\mu})$; $(\beta_1 \beta_2 \dots \beta_\mu)$, $(\beta_{\mu+1} \beta_{\mu+2} \dots \beta_{2\mu})$, $\dots$, $(\beta_{(k-1)\mu+1} \beta_{(k-1)\mu+2} \dots \beta_{k\mu})$. Here we assume $N = k\mu$, if the length of the last group is less than μ , then add the same arbitrary bits at the end of the last groups of both Alice and Bob until its length is μ .

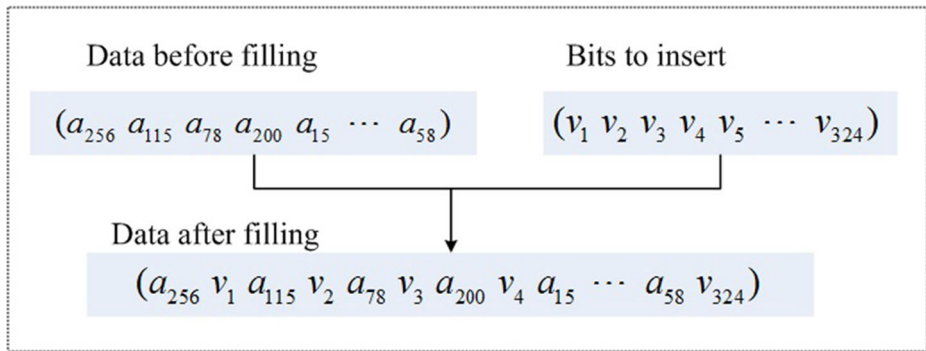


Fig. 3 The bit-filling method

Bit-filling: Bit-filling is to add $n - \mu$ bits to each group, so the length of each group is n , here the dispersed insertion method is used, Fig. 3 is an example of dispersed insertion with $n = 648$, $\mu = 324$, $V = (v_1 v_2 \cdots v_{324})$ is binary sequence generated randomly with equal probability of a zero or a one. By using this method, the error code rates have halved.

In order to provide a good representation of the symbol, after data grouping and bit-filling, Let us suppose that Alice has binary sequence $(x_1 \ x_2 \cdots x_n)$, $(x_{n+1} \ x_{n+2} \cdots x_{2n})$, $\cdots$, $(x_{(k-1)n+1} \ x_{(k-1)n+2} \cdots x_{kn})$, Bob has binary sequence $(y_1 y_2 \cdots y_n)$, $(y_{n+1} \ y_{n+2} \cdots y_{2n})$, $\cdots$, $(y_{(k-1)n+1} \ y_{(k-1)n+2} \cdots y_{kn})$.

For each of the bit filled group, information reconciliation algorithm based on LDPC codes is applied. It includes the above encoding scheme of LDPC and decoding scheme based on sum-product decoding. For each group, if the maximum allowed iteration number of decoder is reached before $S = S'$, it means that the information reconciliation of this group is failure, and then this group is marked error with a special label.

Error groups iterative reconciliation: If the acceptable error correction rate does not reached, Bob will send the information of the marked error groups to Alice, Alice will divide each marked error group into two sub-groups, bit-filling is to add $n/2$ bits to each sub-group. So each sub-group has the length of n , the error code rate has halved again, then repeat the above step 1 to step 5 to start a new round of information reconciliation. Therefore if the error code rate is high, information reconciliation can be implemented repeatedly in order to reduce error code rate or achieve zero error code rate. Alice and Bob are required to adopt interactive communication by adopting this method, while in the later simulation experiment (Fig. 7), we can see that error code rate can be reduced to zero in no more than two rounds of information iterations when QBER approaches the security limit of 11% [29, 31].

Finally, if the maximum rounds of allowed iterative information reconciliation is reached, groups that cannot be corrected will be marked with special label, each group deletes the filling bits, the shared sequence G is used again to recover the initial data $\hat{A} = (\hat{a}_1 \hat{a}_2 \cdots \hat{a}_N)$, then the bit has the special marked error label will be discard. Then both Alice and Bob obtain the same length shared key to work with.

In summary, the block diagram of proposed iterative information reconciliation based on LDPC code is shown in Fig. 4

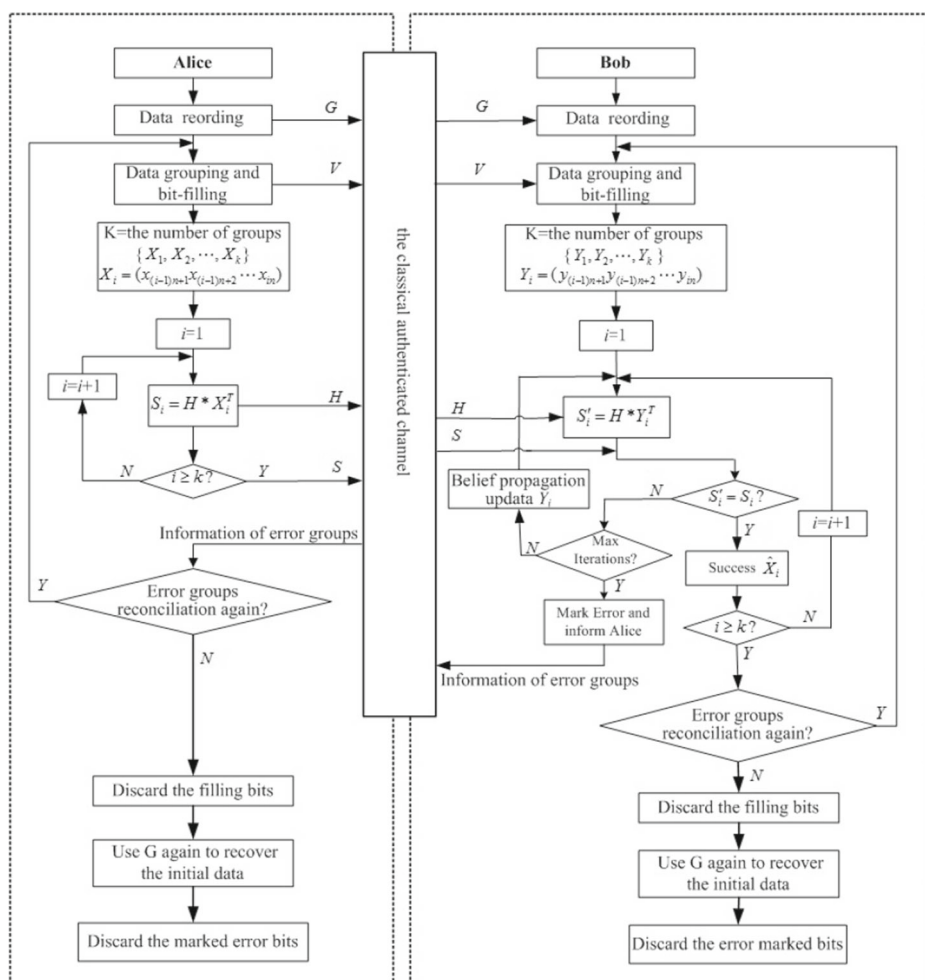


Fig. 4 Block diagram of iterative information reconciliation based on LDPC codes

4 Simulation

In this section, we discuss the experimental performances of the proposed iterative information reconciliation algorithm.

After the quantum transmission, quantum reception and channel parameter estimation, Alice and Bob use iterative information reconciliation algorithm try to make their sifted key identical over the authenticated classical channel. Let us suppose Alice has an N -bit binary string $A = (a_1 a_2 \dots a_N)$, Bob has $B = (b_1 b_2 \dots b_N)$, here if the QBER is higher than a certain value, the eavesdroppers are considered to exist and the communication is discard.

During this part of information reconciliation, one problem of applying LDPC codes is to build the parity-check matrix H , which is beyond the scope of this work. We use the IEEE

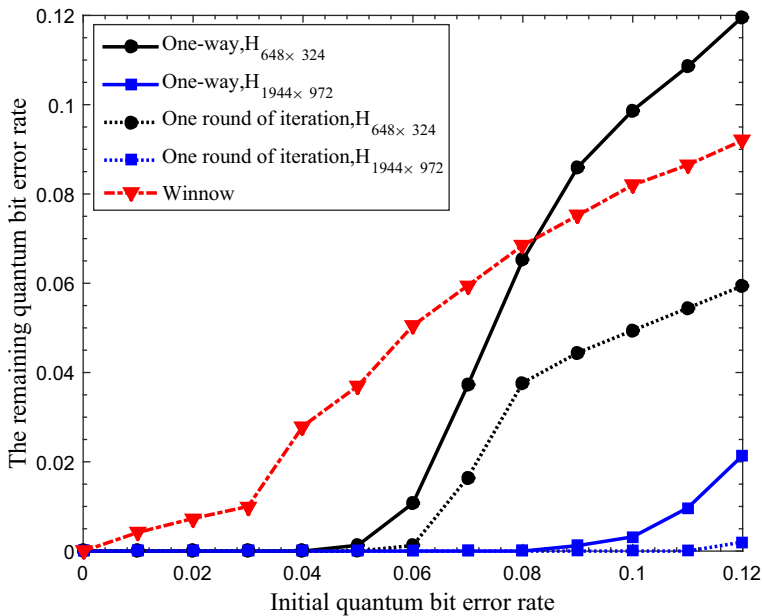


Fig. 5 The information reconciliation capability with different scheme, In order to compare with the one-way scheme, bit filling is not performed before the first round o iterative information reconciliation, the same as in the following figures

802.11n Std to build the parity-check matrix H . Performance has been computed under iterative decoding by using the sum product algorithm, the maximum iteration number of the decoder is set to 100.

Parameters that affect the ability of LDPC codes to locate and correct errors are initial QBER within the sifted key and its distribution, length of the divided group size. The parameters that are taken into account for this experiment are QBER and the length of the divided groups. Here two different data grouping sizes are chosen to examine the error correction probability, because the length of divided groups are related to the LDPC parity-check matrix H , H is a $(n - k) \times n$ matrix, where $n = 648$ (hereinafter referred to as scheme I) and $n = 1944$ (hereinafter referred to as scheme II) are choose, and the code rate R is $1/2$, so the divided group sizes after bit filling are 648 and 1944 respectively, Bit filling is to add $n/2$ filling bits to each group, that is 324 filling bits each group for scheme I , and 972 filling bits each group for scheme II . We set the total length of the sifted key is 100,000 bits.

Figure 5 shows the initial QBER before information reconciliation and the remaining QBER after information reconciliation.

We can see that information reconciliation ability of $H_{1944 \times 972}$ is better than $H_{648 \times 324}$, that is because under certain bit rate conditions, the longer the code length, the stronger the error correction ability. After the first round of iterative information reconciliation the remaining QBER decreases to 0 by using scheme I when the initial QBER is below 8.0%, and the remaining QBER decreases to 0 by using scheme II when the initial QBER approaches the security limit of 11%. The iterative information reconciliation scheme is also better than the Winnow algorithm.

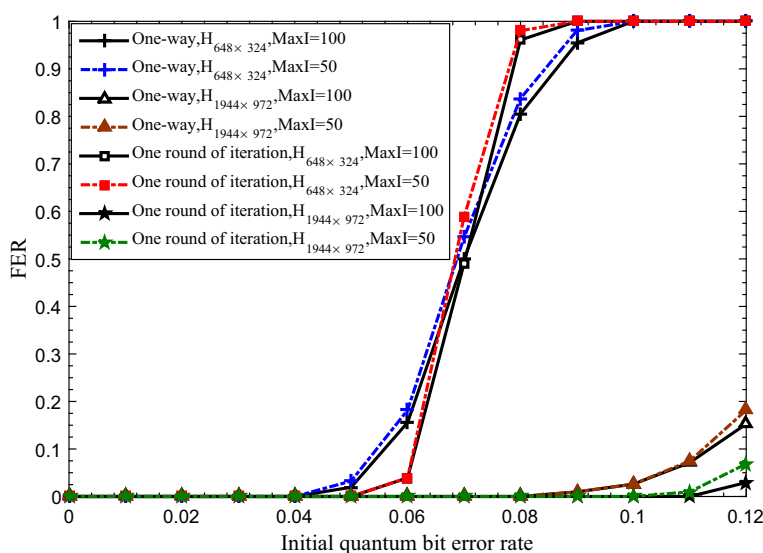


Fig. 6 FER when the maximum iteration number of the LDPC decoder being 50 and 100 respectively

Figure 6 shows FER of the information reconciliation. We can see that the error correction ability with one round of iterative information reconciliation is higher than the one-way reconciliation scheme, that because the QBER is reduced to half in the first round of iteration, the QBER cannot be corrected in the one-way scheme can be further corrected.

As shown in Fig. 6, when the maximum iteration number of LDPC sum-product decoding is set relatively large, the reconciliation capability is improved, this shows that the maximum iteration number of LDPC sum-product decoding also affects the reconciliation capability. If the number is set too low, some groups will be discarded in one-way scheme, because correcting this group requires more iterations than the one set by the system. It is also found that for some groups, no matter how many iterations decoding they are, they cannot be corrected. Therefore, setting the maximum number of LDPC sum-product iterative decoding reasonably cannot only improve the reconciliation ability, but also save the decoding time as much as possible.

Then we consider the second round of iterative information reconciliation, because of bit filling, the initial QBER less than 16% can be corrected by using scheme *I*, and the initial QBER less than 22% can be corrected by using scheme *II*. Generally the QBER over 11% are considered unacceptable for BB84, and the eavesdroppers are considered to exist and the communication will be discard, obviously we can see that QBER can be reduced to zero in no more than two rounds of iteration if the initial QBER is less than 11%.

Figure 7 shows average iteration number of the LDPC decoder when the maximum iteration number of the LDPC decoder is set to 100, although the bit filling scheme increases the number of groups, the number of average iterative decoding decreases significantly.

Then, analyses are provided to study the effects of burst errors in the proposed information reconciliation method. These types of error are removed or mitigated by using our method because of the data reordering and bit filling.

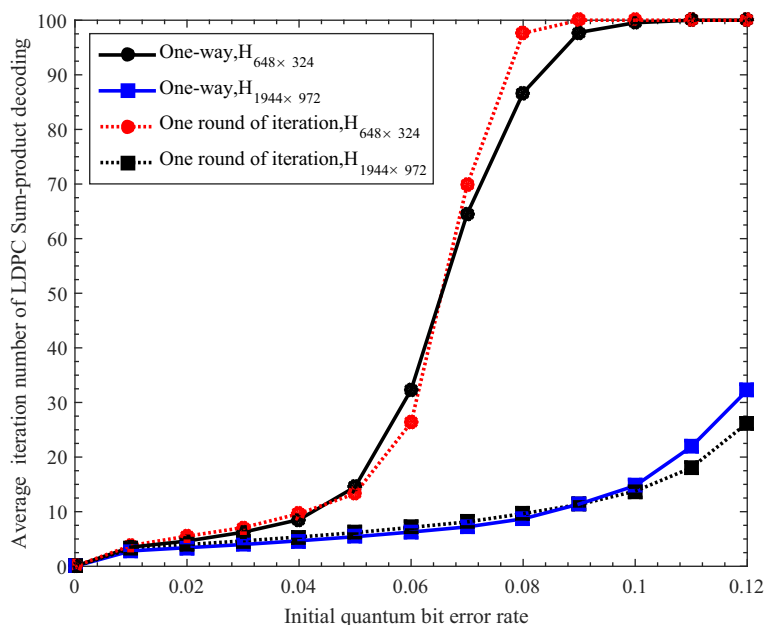


Fig. 7 Average iteration number of LDPC decoder (where the maximum iteration number of the LDPC decoder is set to 100)

5 Conclusion

Due to the problem of high quantum bit error rates in longer distances or in the condition of noisier environments in QKD, in this paper we describe an iterative information reconciliation method based on LDPC codes to improve this problem. Data reordering, data grouping and bit filling are used to improve the error-correcting capability, and reduce the impact of burst errors. If the acceptable error correction rate does not reached, error groups iterative reconciliation are implemented, it is no need to discard the error groups compare with the one-way reconciliation protocol, which can further improve the error correction capability and retain more useful raw data. The simulation experiments have shown that the error correction ability of the iterative information reconciliation scheme proposed in this paper is beyond the results of traditional one-way schemes. Of course the proposed scheme achieves a compromise between reconciliation speed and reconciliation efficiency, but with the advent of 5G, the cost of sacrifice on reconciliation speed will be getting smaller and smaller [32].

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